

Improved Least Lncosh Based Fetal Electrocardiography Extraction in Alpha-Stable Noise

Mengjia Wang[✉], Bo Ni[✉], Jiacheng Zhang[✉], Shengyang Luan[✉], *Member, IEEE*, and Tao Liu[✉]

Abstract—Fetal electrocardiography (FECG) presents an important avenue for continuous fetal monitoring. Nonetheless, effectively extracting FECG from maternal electrocardiogram (MECG) is one considerable challenge due to its weaker amplitude compared to MECG and the non-Gaussian nature of background noise. In this letter, we introduce alpha-stable noise to model the realistic interference due to its high scalability. To improve the accuracy of FECG extraction under impulsive noise (alpha-stable noise with strong impulses), we introduce the least Lncosh algorithm (LLncosh) and the improved Lncosh algorithm (ILL) is proposed based on the Amplitude Hyperbolic Tangent Transformation (AHTT) to optimize the preset parameter. Moreover, Monte Carlo experiments are carried out to investigate the capabilities of LMS-like algorithms and the ILL algorithm for FECG extraction. The results demonstrate that the ILL algorithm outperforms the LMS-like ones with carefully selected parameters, particularly showcasing superior robustness against impulsive noise. This work holds significance both in the theoretical research of adaptive filtering and in its practical application for FECG extraction.

Index Terms—LMS algorithm, LLncosh algorithm, fetal electrocardiogram, alpha-stable noise, impulsive noise.

I. INTRODUCTION

EXTRACTION of a clean fetal electrocardiogram (FECG) is crucial for fetal monitoring. However, FECG is a relatively weak signal, amounting to less than 20% of the maternal electrocardiogram (MECG) in amplitude [1]. In addition, FECG is often completely submerged in electrical interference and

maternal ECG components [2]. These noise sources typically exhibit a pulse characteristic. Hence, it is critical to optimize the elimination of impulsive noises and accurately capture the FECG from MECG.

Adaptive filtering is highly preferred for processing and analyzing the biomedical signal issues mentioned above owing to its low computational complexity and acceptable performance [3]. Traditional least mean square algorithm (LMS) as well as its variants, including normalized LMS (NLMS) and improved proportional normalized LMS (IPNLMS), are widely used [4]. The aforementioned LMS-like algorithms are constructed on the basis of the mean square error (MSE) criterion [5]. However, algorithms based upon the MSE criterion can only achieve optimal estimation under Gaussian assumptions, leading to performance degradation in non-Gaussian noise (impulsive noise) [6]. To improve the performance of algorithms under impulsive noises, alternative criteria beyond MSE have been explored.

In previous studies [7], [8], set-membership filtering is a powerful method against impulsive environments. Recently, a novel cost function called the Lncosh cost function has been successfully developed for processing non-Gaussian noise [9], [10], [11], [12], [13]. The Lncosh cost function does not rely on particular noise distribution assumptions and exhibit better performance in impulsive noise [10]. In [14], the authors pointed out that the Lncosh function is a combination of criteria MSE and mean absolute error (MAE) by adjusting the parameter λ . Therefore, we consider applying the least Lncosh algorithm (LLncosh) derived from the Lncosh cost function to FECG extraction. However, the LLncosh algorithm has a high-dimensional search space and a complex process for a suitable value of the variable parameter λ . To address this issue, we introduce the Amplitude Hyperbolic Tangent Transformation (AHTT) to adjust λ and the corresponding update rule called the improved LLncosh (ILL) algorithm is deduced. Simulation carried out with alpha-stable noise to model the realistic interference [15], the results display the improved performance achieved by the proposed ILL algorithm in FECG extraction.

II. RELATED WORK

A. Signal Model and Adaptive Filtering

A block diagram of FECG extraction using adaptive filters is shown in Fig. 1. The filtered output signal $y(n)$ is calculated via the input signal $x(n)$ and the weight vector $w(n)$, which can be described by:

$$y(n) = w^T(n)x(n), \quad (1)$$

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Mengjia Wang and Bo Ni are with the School of Electronics and Information Engineering, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: wang_mengjia@nuist.edu.cn; 002545@nuist.edu.cn).

Jiacheng Zhang is with the School of Artificial Intelligence, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: zhangjc@nuist.edu.cn).

Shengyang Luan is with the School of Electrical Engineering and Automation, Jiangsu Normal University, Xuzhou 221116, China (e-mail: luan@jsnu.edu.cn).

Tao Liu is with the School of Artificial Intelligence, Nanjing University of Information Science and Technology, Nanjing 210044, China, and also with the Key Laboratory of Pattern Recognition and Intelligent Information Processing, Institutions of Higher Education of Sichuan Province, Chengdu University, Chengdu 610106, China (e-mail: tao.liu@nuist.edu.cn).

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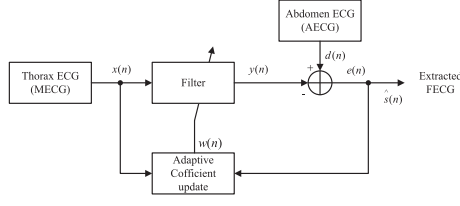


Fig. 1. Block diagram of FECG extraction using adaptive filters.

$$\mathbf{x}(n) = [x(n), x(n-1), \dots, x(n-M+1)]^T, \quad (2)$$

$$\mathbf{w}(n) = [w_1(n), w_2(n), \dots, w_M(n)]^T, \quad (3)$$

where M denotes the order of the filter. The desired input is maternal abdominal ECG (AECG) $d(n)$, which consists of pure FECG $s(n)$ as well as the additive noise $\eta(n)$, so $d(n)$ can be given by:

$$d(n) = s(n) + \eta(n), \quad (4)$$

where $\eta(n)$ includes the maternal ECG component $m(n)$ and other electrical noises $v(n)$. According to relevant studies [16], before superimposing $m(n)$ with $s(n)$, the power of $m(n)$ needs to be reduced. As traditionally done, $m(n)$ is adjusted to 0.2 of the amplitude of MECG [17]. Thus, AECG consequently be described as:

$$d(n) = s(n) + \underbrace{m(n)}_{\text{FECG}} + \underbrace{v(n)}_{\text{Noise}} = \text{FECG} + 0.2\text{MECG} + \text{Noise}. \quad (5)$$

Through the error $e(n)$ between $y(n)$ and $d(n)$, the adaptive filter automatically adjusts $\mathbf{w}(n)$ and minimize the objective function $\mathbf{J}(e(n))$. The estimation of FECG is the output signal $\hat{s}(n)$ [18]. This process is expressed theoretically as follows:

$$\mathbf{w}(n+1) = \mathbf{w}(n) - \mu \nabla \mathbf{J}(e(n)), \quad (6)$$

$$\hat{s}(n) = e(n) = d(n) - y(n). \quad (7)$$

1) *NLMS*: According to stochastic gradient descent (SGD) criterion, the weight vector of LMS algorithm is given by:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + 2\mu_{\text{LMS}}e(n)\mathbf{x}(n). \quad (8)$$

When $\mathbf{x}(n)$ is much greater than $\mathbf{w}(n)$, the gradient term $2\mu_{\text{LMS}}e(n)\mathbf{x}(n)$ becomes the main part of the update. To address this, NLMS algorithm is introduced and derived as [19]:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu_{\text{NLMS}}e(n)\mathbf{x}(n)}{\mathbf{x}^T(n)\mathbf{x}(n) + \delta_{\text{NLMS}}}, \quad (9)$$

where $\mu_{\text{NLMS}} \in (0, 2)$. δ_{NLMS} is a regularisation constant.

2) *IPNLMS*: With further researches, IPNLMS algorithm is proposed and the idea inside is to assign a separate adaptive step size to each filter coefficient [20].

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu_{\text{IPNLMS}}e(n)\mathbf{G}(n)\mathbf{x}(n)}{\mathbf{x}^T(n)\mathbf{G}(n)\mathbf{x}(n) + \delta_{\text{IPNLMS}}}, \quad (10)$$

$$\mathbf{G}(n) = \text{diag}[g_1(n), g_2(n), \dots, g_m(n), \dots, g_M(n)], \quad (11)$$

$$g_m(n) = \frac{(1 - \alpha_{\text{IPNLMS}})}{2M} + \frac{(1 + \alpha_{\text{IPNLMS}})|w_m(n)|}{2M\|\mathbf{w}(n)\| + \lambda_0}. \quad (12)$$

Updating (10), where $\mu_{\text{IPNLMS}} \in (0, 2)$ is a fixed-step parameter, $\mathbf{G}(n)$ is a diagonal matrix regulating the fixed-step of each tap,

δ_{IPNLMS} and λ_0 are tiny positive constants. $\alpha_{\text{IPNLMS}} \in (0, 2)$ is the proportional term to combine the first and second terms on the right-hand side in (12). The parameters $\delta_{\text{IPNLMS}} = 10^{-10}$, $\alpha_{\text{IPNLMS}} = 0.2$, and $\lambda_0 = 10^{-10}$ are used due to the stability indicated by Monte Carlo simulations. While the IPNLMS algorithm demonstrates superior accuracy, it has a higher level of computational complexity.

B. Alpha-Stable Noise

There is no closed-form expression for the alpha-stable noise distribution, which is derived from the generalized central limit theorem (GCLT) [11]. Instead, its characteristic function (CF) is represented as follows [21]:

$$\psi(t) = \exp \{j\theta t - |t|^\alpha \gamma^\alpha [1 - j\beta \text{sgn}(t)\varphi(t, \alpha)]\}, \quad (13)$$

where

$$\varphi(t, \alpha) = \begin{cases} \tan(\pi\alpha/2) & \alpha \neq 1 \\ -2 \ln |t|/\pi & \alpha = 1. \end{cases} \quad (14a)$$

$$(14b)$$

$\alpha \in (0, 2]$ and $\theta \in (-\infty, \infty)$ denote the characteristic exponent and location, respectively. $\beta \in [-1, 1]$ denotes the symmetry parameter. $\gamma \in (0, \infty)$ denotes the dispersion. When $\beta = 0$, the distribution is symmetric, called the symmetric alpha-stable ($S \propto S$) distribution [22], in which case, γ can be fast computed by the given values of α and the generalised signal-to-noise ratio (GSNR) such as:

$$\gamma = (P_{\text{dn}}/10^{\text{GSNR}/10})^{1/\alpha}, \quad (15)$$

where P_{dn} denotes the energy of the desired signal $d(n)$. When $\alpha = 2$, alpha-stable distribution degenerates to the traditional Gaussian distribution. With the four parameters, alpha-stable noise is represented by $S(\alpha, \beta, \gamma, \delta)$.

III. THE IMPROVED LEAST LNCOSH ALGORITHM

A. The Least Lncosh Algorithm (LLncosh)

Literature [14] proposed that the lncosh cost function featuring the natural logarithm of the hyperbolic cosine. Additionally, the lncosh cost function is not reliant on the noise assumptions [23], which is in contrast to the MSE criterion, exhibiting its enhanced robustness in the presence of impulsive noise. The lncosh cost function is described as:

$$\mathbf{J}_{\text{Lncosh}}(e(n)) = \mathbf{E} [\ln(\cosh(\lambda e(n))) / \lambda], \quad \lambda > 0, \quad (16)$$

where $\cosh(\lambda e(n))$ is the hyperbolic cosine function.

According to (6), the weight updating equation $\mathbf{w}(n)$ of the least lncosh algorithm can be derived from SGD:

$$\begin{aligned} \mathbf{w}(n+1) &= \mathbf{w}(n) - \mu_{\text{Lncosh}} \nabla \mathbf{J}_{\text{Lncosh}}(e(n)) \\ &= \mathbf{w}(n) - \mu_{\text{Lncosh}} \frac{\partial \mathbf{J}_{\text{Lncosh}}(e(n))}{\partial \mathbf{w}(n)}, \end{aligned} \quad (17)$$

where

$$\begin{aligned} \frac{\partial \mathbf{J}_{\text{Lncosh}}(e(n))}{\partial \mathbf{w}(n)} &= \frac{1}{\lambda} \frac{\partial \ln(\cosh(\lambda e(n)))}{\partial \mathbf{w}(n)} \\ &= \frac{1}{\lambda} \frac{\partial \ln(\cosh(\lambda e(n)))}{\partial \cosh(\lambda e(n))} \frac{\partial \cosh(\lambda e(n))}{\partial \lambda e(n)} \frac{\partial \lambda e(n)}{\partial \mathbf{w}(n)} \end{aligned}$$

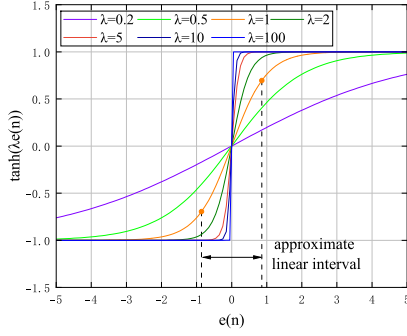


Fig. 2. The behavior of $\tanh(\lambda e(n))$ for different values of λ .

$$\begin{aligned} &= \frac{1}{\lambda} \frac{1}{\cosh \lambda(e(n))} \sinh(\lambda e(n))(-\lambda x(n)) \\ &= -\tanh(\lambda e(n))x(n), \end{aligned} \quad (18)$$

thus,

$$w(n+1) = w(n) + \mu_{\text{Lncosh}} \tanh(\lambda e(n))x(n). \quad (19)$$

The variable form $\lambda(n)$ of parameter λ is introduced in [14] and its expression is given by:

$$\lambda(n) = \lambda_0 \times \frac{1 - \exp(-A_0|e(n)|)}{1 + h \exp(-A_0|e(n)|)}, \quad (20)$$

where parameter A_0 and h control the characteristic of (20).

However, this method increases the search dimensions of the parameters, raising the algorithm's complexity. It often requires presetting the values of parameters λ_0 , A_0 , and h based on experience, which lacks universality. To address this problem, we propose the improved least lncosh algorithm (ILL), incorporating the Amplitude Hyperbolic Tangent Transformation (AHTT) to adjust λ .

B. The Improved Least Lncosh Algorithm (ILL)

Fig. 2 presents the behavior of the function $\tanh(\lambda e(n))$ for different values of λ . It can be observed from Fig. 2 that the function is linear at $e(n) = 0$, as $|e(n)|$ increases, the linearity of $\tanh(\lambda e(n))$ rapidly deteriorates. When $\lambda = 1$, for the real variable $e(n)$, the slope $K(e(n))$ of $\tanh(e(n))$ is:

$$K(e(n)) = \frac{|\tanh''(e(n))|}{1 + [\tanh'(e(n))]^2}, \quad (21)$$

where

$$\begin{cases} \tanh'(e(n)) = 1 - \tanh^2(e(n)), \\ \tanh''(e(n)) = -2 \tanh(e(n))[1 - \tanh^2(e(n))], \end{cases} \quad (22a) \quad (22b)$$

substituting into (21) yields:

$$K(e(n)) = \frac{2|\tanh(e(n))| \operatorname{sech}^2(e(n))}{1 + \operatorname{sech}^4(e(n))}. \quad (23)$$

The maximum value of $K(e(n))$ is calculated through $K'(e(n))$, which is given by:

$$K'(e(n)) = \operatorname{sgn}(e(n)) \frac{8 - \cosh(6e(n)) + 25 \cosh(2e(n))}{8[1 + \operatorname{sech}^4(e(n))]^2}. \quad (24)$$

Set $K'(e(n)) = 0$, we can obtain the roots by numerical computation,

$$e_1(n) = 0, \quad e_2(n) \approx -0.84, \quad e_3(n) \approx 0.84. \quad (25)$$

Namely, $\tanh(e(n))$ is approximately linear when $e(n) \in [-0.84, 0.84]$. As shown in Fig. 2, the approximate linear interval of $\tanh(\lambda e(n))$ extends or compresses as λ varies. Based on this, we introduce the AHTT, which is defined as:

$$\mathbf{T}_H(e(n)) = \tanh(0.84e(n)/H), \quad 0.84/H = \lambda. \quad (26)$$

Relevant researches indicated that the scale parameter H in hyperbolic tangent-like methods is only related to the signal. Depending on the amplitude of the signal we interest, it can be interpreted as an amplitude limiter [24].

In this signal model, the amplitude of the reference input signal $x(n)$ is known. We aim to obtain the FECG signal in $d(n)$ through filtering but cannot initially determine the range of its amplitude. It is worth noting that $d(n)$ contains a component of known amplitude, the maternal ECG component $m(n) = 0.2\text{MECG}$, which is similar to the actual amplitude of the FECG and usually smaller than that of noise. Therefore, the approximate amplitude range of FECG in $d(n)$ can be estimated. Thus, the value of H is given by:

$$H = \max\{|\max\{m(n)\}|, |\min\{m(n)\}|\}. \quad (27)$$

Using the amplitude of $m(n)$ as a reference, the FECG amplitude range is approximated via scale transformation. This keeps parts of $d(n)$ nearing the FECG amplitude falling within the linear range while abnormal amplitudes fall within the limiting range, enhancing FECG extraction accuracy.

The cost function of ILL algorithm is redefined as:

$$\mathbf{J}_{\text{ILL}}(e(n)) = \mathbf{E}[H \ln(\cosh(0.84e(n)/H))/0.84], \quad (28)$$

the derivative of (28) with respect to $w(n)$ is derived as:

$$\begin{aligned} \frac{\partial \mathbf{J}_{\text{ILL}}(e(n))}{\partial w(n)} &= \frac{H}{0.84} \frac{\ln(\cosh(\frac{0.84}{H}e(n)))}{\partial w(n)} \\ &= \frac{H}{0.84} \frac{1}{\cosh(\frac{0.84}{H}e(n))} \sinh(\frac{0.84}{H}e(n)) \left(-\frac{0.84}{H}x(n)\right) \\ &= -\tanh(0.84e(n)/H)x(n), \end{aligned} \quad (29)$$

thus, the formula for updating the coefficients is adjusted to:

$$\begin{aligned} w(n+1) &= w(n) - \mu_{\text{ILL}} \nabla \mathbf{J}_{\text{ILL}}(e(n)) \\ &= w(n) + \mu_{\text{ILL}} H \tanh(0.84e(n)/H)x(n). \end{aligned} \quad (30)$$

C. Convergence Analysis and Complexity

Based on the signal model in Section II, the MECG signal from subject numbered Sub07 in the FECGSYN database (details provided in Section IV) is selected as the input $x(n)$ and $d(n)$ is obtained by (5). Then we perform a convergence analysis of NLMS, IPNLMS and ILL algorithms in FECG extraction. Fig. 3 presents the average of iterative MSE (AMSE) of the three algorithms under two noise backgrounds. In Fig. 3, the ILL algorithm always achieves faster convergence speed than NLMS and IPNLMS, especially in non-Gaussian noise. This indicates that the ILL algorithm has better robustness against impulsive noise. The parameters used in simulation are NLMS ($M = 180$, $\mu_{\text{NLMS}} = 0.1$), IPNLMS ($M = 200$,

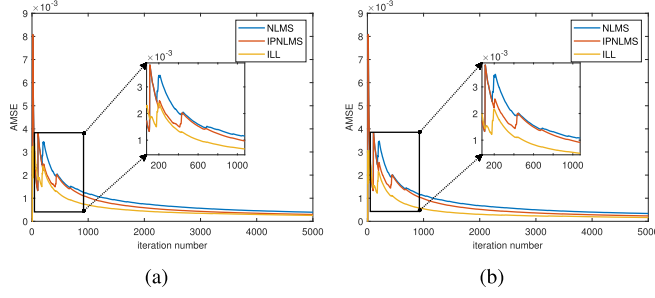


Fig. 3. AMSE of the three algorithms in different noise backgrounds. (a) Gaussian noise with $\alpha = 2.0$, GSNR = 15 dB. (b) non-Gaussian noise with $\alpha = 1.4$, GSNR = 15 dB.

TABLE I
COMPUTATIONAL COMPLEXITY

Algorithms	$\times$	$+$	$\tanh$	$\div$
LMS	2L+1	2L	-	-
NLMS	3L+1	3L	-	1
IPNLMS	6L+4	4L+3	-	3
Lncosh	2L+2	2L	1	1
ILL	2L+3	2L+1	1	1

$\mu_{IPNLMS} = 0.04$, $\alpha_{IPNLMS} = 0.2$) and ILL ($M = 7$, $\mu_{ILL} = 0.03$), which are selected through experiments to achieve relatively optimal filtering. Table I displays the number of required mathematical operations per iteration of each algorithm. Obviously, the computational complexity of NLMS and IPNLMS is significantly higher than that of ILL.

IV. SIMULATION RESULTS

We use the FECGSYN database [25] to provide input data for the abdomen and thorax signals to select the parameters of filters, and evaluate the traces of pregnancy abdomen signals that were identified as zero cases in the database. Traces with a noise level of 6 dB were specifically chosen for evaluation. The simulation consists of varying numbers of $M \in [1, 250]$ and $\mu \in [0.001, 1]$, the average values of the root of the mean square error (RMSE) [18], [26], percentage of the quadratic mean difference (PRD) [18], [26], and signal-to-noise ratio (SNR) [18], [26] are evaluated.

Table II displays the optimal evaluation results of the output response of each algorithm as the noise properties change. From Table II, we can see that with α decreases, the PRD values of NLMS break through 100% in some cases, showing that NLMS cannot adopt to strong impulsive environment. Although IPNLMS has improved its performance compared to NLMS, its computational process is time-consuming. While the ILL algorithm not only has a low computational complexity but also outperforms IPNLMS with SNR improved by 1dB on average, indicating that its ability to suppress impulsive noise is the best among the three algorithms. Fig. 4 presents the graphical responses of NLMS ($M = 90$, $\mu_{NLMS} = 0.2$), IPNLMS ($M = 120$, $\mu_{IPNLMS} = 0.05$, $\alpha_{IPNLMS} = 0.2$) and ILL ($M = 4$, $\mu_{ILL} = 0.01$) in FECG extraction (subject Sub01) when $S(1.4, 0, \gamma, 0)$. The abdominal signal shown in Fig. 4 has MECG and FECG components that are not overlapped, fully overlapped, and partially overlapped, as illustrated by the red solid ellipses 1–3,

TABLE II
OPTIMAL EVALUATION RESPONSE FOR EACH ALGORITHM

Subject	RMSE(dB) / SNR(dB) / PRD(%) under GSNR = 15dB			
	α	NLMS	IPNLMS	ILL
Sub01	2.0	4.34/4.54/64.43	3.20/6.44/49.52	2.89/7.25/46.18
	1.8	4.40/4.49/65.67	3.28/6.35/51.90	2.97/7.17/47.67
	1.6	4.49/4.44/68.86	3.37/6.31/54.32	3.07/7.11/51.20
	1.4	4.85/4.22/87.19	3.77/6.01/72.31	3.55/6.69/70.71
Sub03	2.0	2.29/2.46/95.12	1.41/3.73/77.59	1.09/4.33/72.10
	1.8	2.36/2.46/98.66	1.45/3.79/80.90	1.08/4.46/74.88
	1.6	2.40/2.52/105.40	1.42/3.90/87.23	1.10/4.59/82.41
	1.4	2.87/2.37/150.12	1.99/3.33/88.70	1.65/4.39/83.87
Sub05	2.0	2.49/4.31/70.84	1.72/5.49/59.38	1.42/6.00/55.44
	1.8	2.56/4.30/73.24	1.76/5.53/61.63	1.45/6.07/57.57
	1.6	2.64/4.30/78.83	1.82/5.60/66.99	1.51/6.14/63.16
	1.4	3.07/4.10/110.14	2.28/5.34/98.33	2.07/5.80/97.34
Sub07	2.0	2.92/4.29/71.94	1.74/6.10/54.69	1.06/7.41/46.76
	1.8	2.99/4.23/74.06	1.81/6.07/57.15	1.04/7.53/48.56
	1.6	3.05/4.27/78.95	1.81/6.19/61.2	1.07/7.63/53.55
	1.4	13.49/4.05/108.52	2.26/5.93/90.43	1.63/7.27/85.08

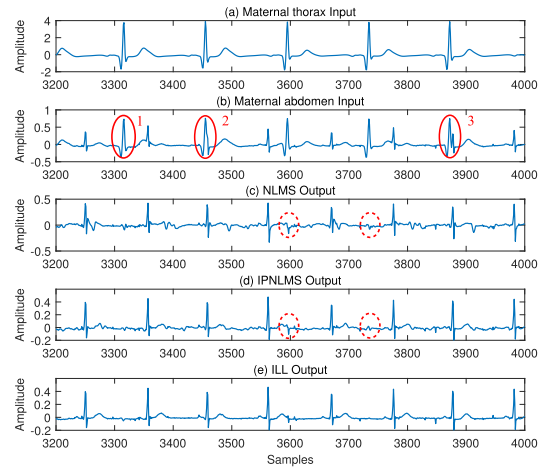


Fig. 4. Graphical response of each algorithm with Sub01 under $S(1.4, 0, \gamma, 0)$.

respectively. The lower performance of NLMS to the other algorithms is evident since it cannot completely eliminate the presence of maternal ECG component $m(n)$ in strong impulsive noise, while IPNLMS exhibits the same issue, as illustrated by the red dotted circles in Fig. 4. Only the ILL algorithm maximally removes $m(n)$ while adopting to the changes in impulse intensity. It offers the best stability and the most effective noise filtering performance.

V. CONCLUSION

In this letter, an ILL algorithm based on the Lncosh algorithm is proposed by introducing the AHTT to reduce the complexity of the parameter λ and enhance its practical application capabilities. We conduct comparative experiments to analyze the performance of the LMS-like algorithms (NLMS, IPNLMS) and the ILL algorithm in FECG extraction under alpha-stable noise. Simulation results show that compared to NLMS and IPNLMS, the proposed algorithm copes best with the abrupt changes generated by the alpha-stable noise with the fastest convergence speed and the best filtering performance.

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